

Erdős Problem #879

EP-879

L1 - TRACTABLE

PARTIALLY SOLVED

Number Theory

ORIGINAL PROBLEM STATEMENT

Call a set $S \subseteq \{1, \dots, n\}$ admissible if $(a, b) = 1$ for all $a \neq b \in S$. Let $G(n) = \max_{S \subseteq \{1, \dots, n\}} \sum_{a \in S} a$ and $H(n) = \sum_{p < n} p + n\pi(n^{1/2})$. Is it true that $G(n) > H(n) - n^{1+o(1)}$? Is it true that, for every $k \geq 2$, if n is sufficiently large then the admissible set which maximises $G(n)$ contains at least one integer with at least k prime factors?

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Archive: <https://www.unsignedmath.com/problems?difficulty=1>

Dataset: <https://huggingface.co/datasets/ulamai/UnsolvedMath>